Coupling Algebras: Gluing, the Square-Zero Cone and the Ring of Contingency Tables

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Abstract

The algebra that carries the geometry of couplings of finite distributions is the *ring of contingency tables*: the Ehrhart ring of a coupling polytope with rational margins. It is generated in degree one, its Hilbert function counts integer tables with scaled margins, and its Proj is the toric variety of the coupling polytope. Its degree-one basis is the sample space of Fisher's exact test, and the basic moves that connect that sample space are integer circulations. For $2 \times m$ tables we give the number of tables in closed form on every chamber of margins: it is a polynomial on each chamber, and crossing a wall changes it by a single binomial term. When the wall cuts a corner of lattice depth d off the polygon, as in the blow-ups of the toric picture, the count drops by $\binom{d+1}{2}$, which is the Riemann–Roch jump of the line bundle. The ring is classical in algebraic statistics; what is new is its place here, as the object that joins the toric geometry of couplings to exact conditional inference.

The other algebra in the programme, the gluing algebra A_f, is much smaller than it looks: $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$, the linear span of the D-algebra endomorphisms of $\mathsf{D}(Y)$ fixing f. The A_∞-functor into the square-zero cone is forced for every mass-preserving linear map but is the mass up to homotopy, and couplings with kernels do not form a double category. These results correct earlier expectations in the programme.

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*Code and checks: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/coupling_algebra. Every statement is checked by `check_coupling_algebra.py` (14 checks, exact arithmetic except where marked).

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1 Introduction

The theory paper [3] linearizes the coupling 2-cells of the Kleisli category of the finite distribution monad. For a distribution f on a finite set, it forms the algebra A_f of matrices whose row and column sums are multiples of f, with the gluing product, and the ideal I_f of circulations. Whiskering by a kernel is then a linear map that is not multiplicative, and its defect is the second Boolean cumulant. Into the square-zero cone of I_f, whiskering is an A_∞-functor with exactly two components [3, Thm. 5.2], also across objects [3, Thm. 10.2]. Two questions were left: why this cone, and whether a larger cone carries more [3, Question 9.4]. A third question, from the attribution paper [2, Open problem 5], asks for a strict structure on the witness layer.

This note has a positive result and a set of corrections. The positive result is that the algebra which carries the coupling geometry is not A_f but a graded commutative one: the ring of lattice points of the coupling polytope, the ring of contingency tables. It joins the toric varieties of [3, 4] to exact conditional inference [11], and its Hilbert function crosses the walls of margins in the same way that the toric surfaces are blown up (proposition 7.3). The ring itself is classical; its role as the meeting point of the two sides is what is new.

The corrections concern the three questions. The cone is homotopy-initial among targets that kill the circulations, but only because it is equivalent to $\mathbb{R}$: the gluing algebra splits off its circulations as a direct factor. The functor into it is forced, and up to homotopy it remembers only the mass. And couplings with kernels do not form a double category.

1.1 For readers of the theory paper

The answer to "why this cone?" is a correction of the strong form of the question, not a confirmation of it. The cone of [3, Thm. 5.2] is the right construction, and nothing about it is a choice: by theorem 5.1, every mass-preserving linear map, whiskering included, becomes an A_∞-functor into it in exactly one way, with $F_2 = \varepsilon b_2$ and no higher components. But the universality one might hope for, that the cone is the initial recipient of the coupling information, is true only in a vacuous form. By theorem 5.2 the cone is quasi-isomorphic to $\mathbb{R}$, and by corollary 5.3 every homotopy invariant of such a functor is a function of masses. So the coupling information of whiskering lives at chain level, in $F_2 = \varepsilon b_2$, and [3, Question 9.4(i)] has a negative answer for homotopy invariants. Whether a free cone carries chain-level information in the higher Boolean cumulants remains open.

1.2 Main results

Main results.

(A) *Structure* (theorem 2.2 and proposition 2.3). $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$. The product coupling $P_f = f \otimes f$ and $e_f = \mathrm{diag}(f) - P_f$ are complementary central idempotents, $I_f = e_f A_f$, and

the mass is the projection onto $\mathbb{R}P_f$. Across objects the linear category of couplings splits as the indiscrete category on $\mathbb{R}$ times the category of circulations.

(B) *Monad and comonad* (propositions 3.1 to 3.3). Couplings of f with itself are the D-algebra endomorphisms of $\mathsf{D}(Y)$ fixing f, gluing is Kleisli composition, and A_f is their linear span. The product coupling is the endomorphism through the zero object $(1, \delta)$, and the mass is the map to it. The tower's context comonad, whose distributive law is the D-algebra structure of the context algebra, is acted on by this monoid, but not affinely at level two, so the action does not extend to A_f.

(C) *Forced functor* (theorem 5.1). For every linear map F_1 that preserves mass, there is exactly one A_∞-functor into the square-zero cone with first component F_1: its second component is the multiplicativity defect, and all higher components vanish.

(D) *Homotopy content* (theorem 5.2 and corollary 5.3). The cone is quasi-isomorphic to $\mathbb{R}$, and the functor composed with this quasi-isomorphism is the mass, a strict algebra map. Since A_f is separable, every A_∞-functor out of it is homotopic to a strict one, and if it makes e_f a boundary, its homotopy class factors through the mass. So no homotopy invariant of such a functor sees contextuality.

(E) *No double category* (proposition 6.1). With couplings as horizontal arrows, kernels as vertical arrows and squares given by pushforward, squares compose horizontally when the middle kernel is a bijection, but not for some constant kernels or some non-injective functions. The defect is a circulation.

(F) *The ring of contingency tables* (theorem 7.1). For margins with common denominator N, the polytope $N \cdot \mathrm{Cpl}(f, g)$ is a lattice polytope with the integer decomposition property. Its Ehrhart ring is generated in degree one, its Hilbert function counts integer tables, and its Proj is the toric variety of the coupling polytope. Its degree-one basis is the sample space of Fisher's exact test, connected by basic moves that are integer circulations.

(G) *Tables across walls* (proposition 7.3). For $2 \times m$ tables the number of tables is given in closed form; it is a polynomial on each chamber of margins, adjacent polynomials differ by one binomial term that vanishes on a strip along the wall, and across a wall that cuts a corner of lattice depth d the count drops by $\binom{d+1}{2}$, the Riemann–Roch jump of a toric blow-up.

1.3 What is not claimed

Earlier drafts of this programme proposed several stronger statements, and each is false. The square-zero cone is homotopy-initial among targets killing I_f, but it is equivalent to $\mathbb{R}$, so initiality says nothing beyond the mass. Its minimal model, in the sense of A_∞-algebras [12], is $\mathbb{R}$ itself. And the witness layer with kernels as vertical maps is not a double category, strict or lax. The gluing algebra is also not a coordinate ring, since it is not commutative, and $A_f/I_f \cong \mathbb{R}$ is not the cohomology ring of the toric manifold, whose dimension is the number of vertices of the polytope. The monadic reading also has limits (proposition 3.3 and remark 3.4): the barycenter is not the mass projection, there is no ideal filtration matching the precision ladder, and the multiplicativity defect is not a Hochschild class.

1.4 Relation to the other papers

This note uses the definitions of [3, §§2, 5, 10] and of the coupling 2-cells of [2, §7]. From the tower paper [6] it uses only the context comonad, its distributive law and the stochastic change of base, in section 3. It does not use the obstruction tower, the empirical material of [5], or the toric computations of [4], except to name the variety in theorem 7.1 and to compare walls in proposition 7.3.

2 The gluing algebra

Let f be a full-support distribution on $Y = \{1, \dots, n\}$, and $D = \operatorname{diag}(f)$.

Definition 2.1. $A_f = \{a \in \mathbb{R}^{n\times n} : a\mathbf{1} = s\,f,\ \mathbf{1}^{\mathsf{T}}a = s\,f^{\mathsf{T}} \text{ for some } s = s(a) \in \mathbb{R}\}$, with product $a \cdot b = aD^{-1}b$ and unit D. The *mass* is $s(a)$, and $I_f = \ker s$ is the space of *circulations*.

Couplings of f with itself are the elements with $a \geq 0$ and $s = 1$, and their product is the gluing of [3, Thm. 2.3]. Write $P_f = ff^{\mathsf{T}}$, the product coupling, and $e_f = D - P_f$.

Theorem 2.2 (Structure of the gluing algebra)**.** *(i) P_f and e_f are idempotents with $P_f + e_f = D$ and $P_f \cdot e_f = 0$, and both are central.*

(ii) $P_f \cdot a = a \cdot P_f = s(a)P_f$ and $e_f \cdot a = a - s(a)P_f$. Hence $A_f = \mathbb{R}P_f \oplus I_f$ as algebras, $I_f = e_f A_f$ is a unital algebra with unit e_f, and s is the projection onto the first factor.

(iii) With $v = (\sqrt{f_y})_y$, the map $\varphi(a) = D^{-1/2}aD^{-1/2}$ is an isomorphism of A_f onto $\{M : Mv = \lambda v,\ v^{\mathsf{T}}M = \lambda v^{\mathsf{T}}\} = \mathbb{R}\,vv^{\mathsf{T}} \oplus \operatorname{End}(v^{\perp})$. So $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$, a semisimple algebra, and $\dim I_f = (n-1)^2$.

Proof. (ii) $P_f D^{-1} a = f(\mathbf{1}^{\mathsf{T}}a) = s(a)ff^{\mathsf{T}}$, and symmetrically. Then $e_f \cdot a = a - s(a)P_f$, which has mass 0. (i) follows from (ii) with $s(P_f) = 1$ and $s(e_f) = 0$, using $f^{\mathsf{T}}D^{-1}f = \sum_y f_y = 1$. (iii) $\varphi(aD^{-1}b) = \varphi(a)\varphi(b)$ and $\varphi(D) = 1$. The conditions $a\mathbf{1} = sf$ and $\mathbf{1}^{\mathsf{T}}a = sf^{\mathsf{T}}$ become $\varphi(a)v = sv$ and $v^{\mathsf{T}}\varphi(a) = sv^{\mathsf{T}}$, since $D^{1/2}\mathbf{1} = v$ and $D^{-1/2}f = v$. If $Mv = \lambda v$ and $v^{\mathsf{T}}M = \lambda v^{\mathsf{T}}$, then M maps $v^{\perp}$ into itself, so M is λvv^{T} plus an endomorphism of $v^{\perp}$; and $\varphi(P_f) = vv^{\mathsf{T}}$. □

So the algebra is as simple as it could be, and none of the coupling geometry is visible in it: the couplings are the nonnegative elements of mass one, and nonnegativity is not an algebraic condition here. The product coupling, the Birch point of the independence model and the Kähler class of [3, §6], appears as the central idempotent that projects onto the masses.

Several objects. Let $\mathcal{L}$ be the linear category of [3, Def. 10.1]: objects are full-support distributions on finite sets, $\mathcal{L}(f, g)$ consists of the matrices with row sums sf and column sums sg, and composition is $a \cdot b = a\,\operatorname{diag}(g)^{-1}b$.

Proposition 2.3 (Splitting)**.** *Every $a \in \mathcal{L}(f, g)$ is uniquely $a = s(a)\,fg^{\mathsf{T}} + x$ with x a circulation, and*

$$\left(s\,fg^{\mathsf{T}} + x\right) \cdot \left(t\,gk^{\mathsf{T}} + y\right) = st\,fk^{\mathsf{T}} + x \cdot y.$$

So each hom-space of $\mathcal{L}$ is the direct sum of $\mathbb{R}$ and the circulations, compatibly with composition: $\mathcal{L}$ is the componentwise sum of the linearized indiscrete category, with every hom-space $\mathbb{R}$, and the category of circulations, whose identity at f is e_f.

Proof. $fg^{\mathsf{T}}\operatorname{diag}(g)^{-1}y = f(\mathbf{1}^{\mathsf{T}}y) = 0$ for a circulation y, and $x\,\operatorname{diag}(g)^{-1}gk^{\mathsf{T}} = (x\mathbf{1})k^{\mathsf{T}} = 0$; the remaining term is $st\,f(\mathbf{1}^{\mathsf{T}}\operatorname{diag}(g)^{-1}g)k^{\mathsf{T}} = st\,fk^{\mathsf{T}}$. □

3 The monad and the comonad

The gluing algebra is defined by matrices, but it comes from the distribution monad D. This section says how, and how far the context comonad of the tower paper [6] reaches into it. Write $K^{\#}\colon \mathsf{D}(Y) \to \mathsf{D}(Y)$, $K^{\#}(\rho) = K^{\mathsf{T}}\rho$, for the Kleisli extension of a kernel $K\colon Y \to \mathsf{D}(Y)$, written as a stochastic matrix. The Kleisli extensions are exactly the morphisms of D-algebras $\mathsf{D}(Y) \to \mathsf{D}(Y)$ between free algebras, that is, the affine self-maps of the simplex. Let

$$\mathrm{End}(f) = \{K\colon Y \to \mathsf{D}(Y) \ : \ K^{\#}f = f\},$$

the monoid, under Kleisli composition, of D-algebra endomorphisms of $\mathsf{D}(Y)$ that fix the point f.

Proposition 3.1 (Couplings are invariant kernels)**.** *(i) The map $\alpha \mapsto D^{-1}\alpha$ is an isomorphism of monoids from* $\mathrm{Cpl}(f,f)$ *under gluing onto* $\mathrm{End}(f)$ *under Kleisli composition, with the unit D going to the identity kernel. Its linear extension $a \mapsto D^{-1}a$ is an isomorphism of A_f onto the algebra of matrices M with $M\mathbf{1} = s\mathbf{1}$ and $M^{\mathsf{T}}f = sf$, and A_f is the linear span of the couplings, so of* $\mathrm{End}(f)$*.*

(ii) Let $\mathcal{P}$ be the category whose objects are the full-support probability spaces (Y,f), that is, the objects of the coslice of the Kleisli category under the one-point set, restricted to full support, and whose morphisms are the kernels that push the first distribution to the second. The one-point space $(1,\delta)$ is a zero object of $\mathcal{P}$. Under (i), $\mathcal{P}(f,g) \cong \mathrm{Cpl}(f,g)$, and $\mathcal{L}$ is the category obtained from $\mathcal{P}$ by taking linear spans of the hom-sets.

(iii) The morphism of $\mathcal{P}$ that factors through the zero object is the constant kernel $y \mapsto g$, which corresponds to the product coupling fg^{T}. So the first summand of proposition 2.3 is spanned by the maps through the zero object, and P_f is the endomorphism of (Y,f) through it.

(iv) The mass is the pushforward along the unique morphism $(Y,f) \to (1,\delta)$: for this kernel ! the product action of section 4 gives $\pi_!(a) = \mathbf{1}^{\mathsf{T}}a\mathbf{1} = s(a)$, and it is multiplicative.

Proof. (i) If $\alpha \in \mathrm{Cpl}(f,f)$ then $K = D^{-1}\alpha$ is nonnegative with $K\mathbf{1} = D^{-1}f = \mathbf{1}$ and $K^{\mathsf{T}}f = \alpha^{\mathsf{T}}\mathbf{1} = f$; conversely $\alpha = DK$ for $K \in \mathrm{End}(f)$ is a coupling. Gluing is $\alpha D^{-1}\beta = D(D^{-1}\alpha)(D^{-1}\beta)$, which is Kleisli composition of the kernels in matrix form. The linear statement is the same computation without positivity. The couplings span A_f because P_f has positive entries, so $P_f + tx$ is a coupling for every circulation x and small t; the span therefore contains $\mathbb{R}P_f + I_f = A_f$. (ii) A morphism $(1,\delta) \to (Y,f)$ is a kernel $1 \to \mathsf{D}(Y)$ sending δ to f, so it is f itself; a morphism $(Y,f) \to (1,\delta)$ is the unique kernel to the one-point set. The isomorphism $\mathcal{P}(f,g) \cong \mathrm{Cpl}(f,g)$ is (i) with two objects, and composition in $\mathcal{L}$ is the linear extension of composition in $\mathcal{P}$. (iii) The composite $Y \to 1 \to \mathsf{D}(Y)$ is $y \mapsto g$, with matrix $\mathbf{1}g^{\mathsf{T}}$, and $D\,\mathbf{1}g^{\mathsf{T}} = fg^{\mathsf{T}}$. (iv) $\mathbf{1}^{\mathsf{T}}aD^{-1}b\mathbf{1} = s(a)\,f^{\mathsf{T}}D^{-1}f\,s(b) = s(a)s(b)$. □

So the gluing algebra is the linearized endomorphism monoid of a point of a free D-algebra, and the structure theorem reads categorically. The central idempotent P_f is the endomorphism through the zero object, the mass is the linearized map to the zero object, and the circulations are the complementary summand. The barycenter $\mu_Y\colon \mathsf{D}^2(Y) \to \mathsf{D}(Y)$ plays no role in the splitting: the mass line comes from the zero object, not from the monad multiplication.

The comonad. The tower paper builds, for a finite base B, the context algebra $A_B = \mathsf{D}B \times \mathsf{D}^2B \times \mathsf{D}^3B \times \mathrm{Sh}(B)$ with the levelwise barycenter β_B, which makes it a product of free D-algebras [6, §9]. The comonad $\mathsf{C}^B(Y) = Y \times A_B$ is the product (reader) comonad with parameter A_B, and the

mixed distributive law $\lambda^B\colon \mathsf{D}\mathsf{C}^B \Rightarrow \mathsf{C}^B\mathsf{D}$ of [6, Thm. 9.8] is $\lambda^B_Y(M) = (\mathsf{D}(\mathrm{pr}_1)M,\ \beta_B(\mathsf{D}(\mathrm{pr}_2)M))$. Mixed distributive laws correspond to liftings of the comonad to the Eilenberg–Moore category [13], and the lifting here sends a D-algebra (X,ξ) to the product algebra $(X,\xi)\times(A_B,\beta_B)$. So the law is exactly the statement that A_B is a D-algebra. The coalgebras of C^B are the sets over A_B, and the cofree coalgebra on X is $X\times A_B$ [6, Props. 13.1, 13.2]. A kernel $k\colon B\to\mathsf{D}(B')$ acts by the D-algebra map $A_k = k^{\#}\times\mathsf{D}(k^{\#})\times\mathsf{D}^2(k^{\#})\times\mathrm{Sh}(k)$, with $A_{h\circ k} = A_hA_k$ [6, Prop. 9.14].

Proposition 3.2 (The comonad sees the monoid, not the algebra)**.** *Take $B = Y$, and let $A^f_Y\subseteq A_Y$ be the contexts whose level-one component is f.*

(i) $K\mapsto A_K$ is an action of the monoid $\mathrm{End}(f)$ on A^f_Y, by D-algebra maps. It is trivial on the level-one component.

(ii) On level one, $K\mapsto K^{\#}$ is affine in K. On level two it is not: for $K_1,K_2\in\mathrm{End}(f)$ with $\bar K = \frac12(K_1+K_2)$ and a context with level-two component Π, in general $\mathsf{D}(\bar K^{\#})\Pi\neq\frac12\mathsf{D}(K_1^{\#})\Pi+\frac12\mathsf{D}(K_2^{\#})\Pi$.

(iii) Hence the action does not extend to a linear action of A_f on the span of the contexts, compatible with proposition 3.1(i). The algebra A_f linearizes the monad side only; the comonad side sees the monoid $\mathrm{End}(f)$.

Proof. (i) A_K is a D-algebra map and $A_{K_2\circ K_1} = A_{K_2}A_{K_1}$ by [6, Prop. 9.14], and $K^{\#}f = f$. (ii) $K^{\#}\rho = K^{\mathsf{T}}\rho$ is linear in K. For the second statement take $\Pi = \frac12\delta_{\rho_1}+\frac12\delta_{\rho_2}$ with $\frac12(\rho_1+\rho_2) = f$. Then $\mathsf{D}(\bar K^{\#})\Pi$ is supported on the two points $\bar K^{\#}\rho_i$, while the average of the $\mathsf{D}(K_j^{\#})\Pi$ is supported on the four points $K_j^{\#}\rho_i$, which are distinct for generic K_1,K_2. This is the phenomenon behind the failure of the shift law and of the level separation in [6, Thm. 8.3, Ex. 10.6]: averaging inside a level is not averaging of levels. (iii) A linear action of the span A_f would restrict to an affine map on the convex set $\mathrm{End}(f)$, contradicting (ii). □

Proposition 3.3 (Limits of the monadic reading)**.** *(i) $D\cdot\mathrm{End}(f) = \{a\in A_f : a\geq 0,\ s(a) = 1\} = \mathrm{Cpl}(f,f)$ is a proper subset of A_f that spans it. It is a convex set closed under gluing, not a subspace.*

(ii) The two-sided ideals of A_f are exactly 0, $\mathbb{R}P_f$, I_f and A_f, and every nonzero circulation generates I_f. So the maximal chains of ideals are $0\subsetneq\mathbb{R}P_f\subsetneq A_f$ and $0\subsetneq I_f\subsetneq A_f$, and there is no chain $0\subsetneq J_1\subsetneq J_2\subsetneq A_f$.

(iii) $HH^n(A_f,M) = 0$ for every A_f-bimodule M and every $n\geq 1$. For a mass-preserving linear $F_1\colon A_f\to A_{f'}$, the rule $a\cdot x = F_1(a)\cdot x$ makes $I_{f'}$ a left A_f-module if and only if $b_2 = 0$.

Proof. (i) is proposition 3.1(i); the set is not closed under negation. (ii) By theorem 2.2(iii), $A_f\cong\mathbb{R}\times M_{n-1}(\mathbb{R})$, a product of two simple algebras, whose two-sided ideals are the four products of 0 or the whole factor. A nonzero $x\in I_f$ lies in the simple factor I_f, so it generates I_f. (iii) A_f is a product of matrix algebras over $\mathbb{R}$, hence separable, so it is projective as a bimodule over itself and its Hochschild cohomology vanishes in positive degrees. The module axiom $F_1(ab)\cdot x = F_1(a)\cdot(F_1(b)\cdot x)$ says $b_2(a,b)\cdot x = 0$ for all $x\in I_{f'}$; taking $x = e_{f'}$, the unit of $I_{f'}$, gives $b_2(a,b) = 0$, since $b_2(a,b)\in I_{f'}$. □

So b_2 satisfies the Hochschild cocycle identity only formally (the arity-three equation of theorem 5.1): there is no bimodule in which it is a cocycle unless it vanishes, and no Hochschild class could carry it anyway.

Remark 3.4 (What the monadic reading does not give). (a) *The barycenter is not the mass projection.* μ_Y is a map $\mathsf{D}^2(Y) \to \mathsf{D}(Y)$, and the projection onto $\mathbb{R}P_f$ is proposition 3.1(iv), induced by the map to the zero object.

(b) *The precision ladder is not an ideal filtration.* The ladder of [2, Thm. 3.5] has three nested rungs, in general distinct, and by proposition 3.3(ii) A_f has no chain of three distinct nonzero ideals. The ladder lives in the 1-cell layer. A decomposition of p with exact sources is a weight vector $(\lambda_1, \dots, \lambda_k, \mu)$ and an unmodeled law U with $p = k^{\#}(\lambda, \mu)$ for the kernel $k\colon i \mapsto A_i$, $u \mapsto U$; so the attribution set is a union, over U, of fibres over p of the Kleisli extension of a kernel, cut by the budget $\mu \le \bar{\mu}$ [2, Def. 3.1, Thm. 3.2]. Sources known by their supports enlarge the set of admissible kernels. The rungs grow because more kernels are admitted and the budget grows, which is a statement about fibres of a 1-cell, not about the gluing of 2-cells.

(c) *No explanation of non-unification.* The splitting is internal to the coupling layer. The non-unification results of [5] concern different coefficient systems, and nothing here maps them to A_f.

By proposition 3.3(i), it is in passing from the monoid to its span that positivity, and with it contextuality, is lost (corollary 5.3).

4 How kernels act

A kernel $h\colon Y \to \mathsf{D}(Z)$, written as a stochastic matrix, acts on matrices in two ways [3, Def. 10.1]. The *product action* $\pi_h(a) = h^{\mathsf{T}} a h$ runs h independently on both coordinates. The *whiskering* w_h copies h on equal pairs and runs it independently on unequal pairs, which is the postcomposition of the witness layer [2, Thm. 7.6]. Both preserve mass and push margins forward.

Proposition 4.1 (The actions in split coordinates)**.** *Write h_y for the row of h at y.*

(i) $\pi_h(fg^{\mathsf{T}}) = (h^{\mathsf{T}} f)(h^{\mathsf{T}} g)^{\mathsf{T}}$*, and π_h maps circulations to circulations. So π_h respects the splitting.*

(ii) *For every a, $w_h(a) = \pi_h(a) + \sum_y a_{yy}\big(\operatorname{diag}(h_y) - h_y h_y^{\mathsf{T}}\big)$, and each term of the sum is a circulation. In particular $w_h(P_f) = P_{h^{\mathsf{T}} f} + c_h(f)$ with the* copy correction $c_h(f) = \sum_y f_y^2\big(\operatorname{diag}(h_y) - h_y h_y^{\mathsf{T}}\big)$.

(iii) *The multiplicativity defect of the whiskering on the product coupling is $b_2(P_f, P_f) = w_h(P_f \cdot P_f) - w_h(P_f) \cdot w_h(P_f) = c_h(f) - c_h(f) \cdot c_h(f)$.*

Proof. (i) is direct, and π_h of a matrix with zero row and column sums has zero row and column sums because h is stochastic. (ii) The two actions agree off the diagonal of a; on an equal pair (y, y) copying replaces $a_{yy} h_y h_y^{\mathsf{T}}$ by $a_{yy} \operatorname{diag}(h_y)$, and $\operatorname{diag}(h_y) - h_y h_y^{\mathsf{T}}$ has zero row and column sums. The diagonal of P_f is $(f_y^2)_y$. (iii) $P_f \cdot P_f = P_f$, and $P_{h^{\mathsf{T}} f}$ annihilates circulations by theorem 2.2(ii). □

So the whiskering mixes the two factors of theorem 2.2, and the copy correction, a weighted sum of the covariance matrices of the rows of h, measures by how much. The product action respects the factors but does not preserve the unit, since $\pi_h(D) = h^{\mathsf{T}} D h \neq \operatorname{diag}(h^{\mathsf{T}} f)$ unless h is deterministic. This is the trade-off of [3, Thm. 10.2] seen in coordinates.

5 The square-zero cone

Let f' be a distribution and $C = A_{f'} \oplus \varepsilon I_{f'}$, with $A_{f'}$ in degree 0 and $\varepsilon I_{f'}$ in degree -1. The product is $a \cdot \varepsilon x = \varepsilon(a \cdot x)$, $\varepsilon x \cdot a = \varepsilon(x \cdot a)$ and $\varepsilon x \cdot \varepsilon y = 0$, which is associative and unital because $I_{f'}$ is a two-sided ideal, and the differential is $m_1(\varepsilon x) = x$, which satisfies the Leibniz rule. So C is a differential graded algebra, whose underlying complex is the mapping cone of $I_{f'} \hookrightarrow A_{f'}$. We use the sign conventions of [12].

Theorem 5.1 (The functor is forced)**.** *Let $F_1 \colon A_f \to A_{f'}$ be linear with $s(F_1 a) = s(a)$. Then there is exactly one A_∞-functor $F \colon A_f \to C$ with first component F_1. Its components are $F_2 = \varepsilon\, b_2$, with $b_2(a, b) = F_1(a \cdot b) - F_1(a) \cdot F_1(b)$, and $F_n = 0$ for $n \geq 3$.*

Proof. The component F_n has degree $1 - n$. For $n \geq 3$ the cone has nothing in that degree, so $F_n = 0$. The arity-two equation reads $m_1 F_2(a, b) = b_2(a, b)$. Since F_1 preserves mass, $b_2(a, b)$ has mass $s(a)s(b) - s(a)s(b) = 0$, so it lies in $I_{f'}$. Since F_2 has degree -1 it lies in $\varepsilon I_{f'}$, where m_1 is injective, and so $F_2 = \varepsilon b_2$ is the only solution. The arity-three equation is $F_1 a \cdot b_2(b, c) - b_2(ab, c) + b_2(a, bc) - b_2(a, b) \cdot F_1 c = 0$. Expanding b_2, the terms $F_1(abc)$ and $F_1 a\, F_1 b\, F_1 c$ cancel, using only linearity of F_1 and associativity. In arity four the only surviving term is a product $F_2 \cdot F_2 \in \varepsilon^2 I_{f'} = 0$, and in higher arity every term contains some F_k with $k \geq 3$. □

Whiskering and the product action are two instances, and so are their convex combinations. The cone is therefore not specific to whiskering: it is the receptacle of every mass-preserving linear map, which it turns into an A_∞-functor by recording the multiplicativity defect.

Theorem 5.2 (Homotopy content)**.** *(i) $H^0(C) = A_{f'}/I_{f'} \cong \mathbb{R}$ and $H^{-1}(C) = 0$, so the map $q(a + \varepsilon x) = s(a)$ is a quasi-isomorphism $C \to \mathbb{R}$ of differential graded algebras. In particular the minimal model of C is $\mathbb{R}$.*

(ii) For the functor F of theorem 5.1, $q \circ F$ is the mass $s \colon A_f \to \mathbb{R}$, a strict algebra map.

(iii) Let $G \colon A_f \to E$ be any A_∞-functor into an A_∞-algebra E such that $G_1(e_f)$ is a boundary. Then $G_1(x)$ is a boundary for every circulation x, and $H(G_1)$ factors through the mass.

Proof. (i) m_1 maps $\varepsilon I_{f'}$ isomorphically onto $I_{f'}$, and $A_{f'}/I_{f'} \cong \mathbb{R}$ by theorem 2.2. The map q is multiplicative because s is and because q vanishes on $\varepsilon I_{f'}$. A dg algebra with cohomology $\mathbb{R}$ in degree 0 has minimal model $\mathbb{R}$. (ii) $q(F_1 a) = s(F_1 a) = s(a)$ and $q(F_2) = 0$. (iii) An A_∞-functor induces an algebra map on homology. For a circulation x, $x = e_f \cdot x$ by theorem 2.2(ii), so $H(G_1)(x) = H(G_1)(e_f)\, H(G_1)(x) = 0$. □

Corollary 5.3. *Let $G \colon A_f \to E$ be an A_∞-functor such that $G_1(e_f)$ is a boundary. Then G is homotopic, after passing to a minimal model of E, to a strict algebra map that factors through the mass s. So every homotopy invariant of G is a function of masses alone.*

Proof. By theorem 2.2, A_f is a product of matrix algebras over $\mathbb{R}$, hence separable, so its Hochschild cohomology with coefficients in any bimodule vanishes in positive degrees. The obstructions to removing the higher components of an A_∞-functor out of A_f, after passing to a minimal model of the target, lie in these groups, so G is homotopic to a strict map. That map is $H(G_1)$, which factors through s by theorem 5.2(iii). □

In particular no such invariant detects contextuality, which is an obstruction to positivity: signed measures always glue [5, Prop. 35.1], and the parity horns of [3, Thm. 4.3] are compatible families

with contextual fraction $1 - 2^{2-n}$ that a linear invariant cannot distinguish from noncontextual ones.

So the coupling information of the A_∞-functor lives at chain level, in $F_2 = \varepsilon b_2$, and nowhere in its homotopy type. This settles [3, Question 9.4(i)] in the negative for homotopy-invariant information. Whether the free cone carries chain-level higher components given by the higher Boolean cumulants remains open (section 8).

6 Couplings with kernels do not form a double category

Take full-support probability spaces (Y, p) as objects, couplings $\alpha \in \mathrm{Cpl}(p, q)$ as horizontal arrows composed by gluing, and kernels k with $k^{\mathsf{T}} p = p'$ as vertical arrows. A square with horizontal arrows $\alpha \in \mathrm{Cpl}(p, q)$ and $\alpha' \in \mathrm{Cpl}(p', q')$ and vertical kernels k, l exists when $\alpha' = k^{\mathsf{T}} \alpha l$; squares are a property, so the structure would be a thin double category. Squares compose vertically, because π is functorial in the kernel. For them to compose horizontally, gluing must commute with pushforward along the middle kernel.

Proposition 6.1 (Interchange)**.** *Let $\alpha \in \mathrm{Cpl}(p, q)$, $\beta \in \mathrm{Cpl}(q, r)$, and kernels k, l, m with l acting on the middle space. The identity*

$$k^{\mathsf{T}}(\alpha \cdot \beta)m = (k^{\mathsf{T}} \alpha l) \cdot (l^{\mathsf{T}} \beta m),$$

where the right side glues over $l^{\mathsf{T}} q$, holds whenever l is a bijection, and more generally whenever l is a function such that on each of its fibres the conditional law of the first coordinate given the middle one, or that of the last coordinate given the middle one, is constant. It fails for some constant kernels l and for some non-injective functions l, so squares are not closed under horizontal composition. The difference of the two sides is always a circulation.

Proof. Since k and m act on the outer coordinates, it suffices to take them to be identities. For a function l write y' for a fibre and $q(y')$ for its mass. With $\alpha_{\cdot y}$ the columns of α and $\beta_{y\cdot}$ the rows of β, the two sides are

$$\sum_{y'} \sum_{y \in y'} \frac{\alpha_{\cdot y} \beta_{y\cdot}}{q_y} \quad \text{and} \quad \sum_{y'} \frac{1}{q(y')} \Big(\sum_{y \in y'} \alpha_{\cdot y} \Big) \Big(\sum_{y \in y'} \beta_{y\cdot} \Big).$$

If $\alpha_{\cdot y} = q_y c_{y'}$ for $y \in y'$, both fibre terms equal $c_{y'} \sum_{y \in y'} \beta_{y\cdot}$, and symmetrically for β. A bijection has singleton fibres. For a counterexample with a function, let $p = q = r$ be uniform on $\{0, 1\}$, let $\alpha = \beta$ be the diagonal coupling, let k, m be identities and let l collapse $\{0, 1\}$ to a point. The left side is the diagonal coupling, since gluing two copies of the identity gives the identity. On the right, $k^{\mathsf{T}} \alpha l$ and $l^{\mathsf{T}} \beta m$ are product couplings with a one-point middle space, so their gluing is the product coupling $\frac{1}{4} \mathbf{1}\mathbf{1}^{\mathsf{T}}$. For a genuinely stochastic kernel, take instead l with both rows uniform on $\{0, 1\}$: then αl and $l^{\mathsf{T}} \beta$ are again product couplings, so the right side is still $\frac{1}{4} \mathbf{1}\mathbf{1}^{\mathsf{T}}$. Both sides have the margins $k^{\mathsf{T}} p$ and $m^{\mathsf{T}} r$, so their difference is a circulation. □

So the structure is a double category, strict and thin, when the vertical arrows are restricted to bijections. The sufficient condition says that interchange holds when the middle map merges only values that are interchangeable for one of the outer variables, that is, when $l(Y)$ is sufficient for Y with respect to X or to Z. With all kernels, horizontal composition of squares is not defined in general, and since squares are a property there are no comparison cells that could make it lax. The defect is a circulation of the same kind as the whiskering defect, so a double-category formulation does not remove the obstruction of [2, Thm. 7.6(v)].

Remark 6.2 (Laxness elsewhere in the programme). Support policies and floor summaries on credal sets are compared along kernels by inclusions, and floor summaries form a normal oplax functor into posets [6, Prop. 10.24]. That laxness belongs to a functor on the Kleisli category with ordered fibres [3, Prop. 3.6], not to the witness layer, so it is consistent with the statement above that the witness layer is not a double category, strict or lax. Cell floors act on the ring of contingency tables only by translating the margins: the floored polytope is a translate of $\mathrm{Cpl}(f', g')$ for the reduced margins [3, Prop. 6.9].

7 The ring of contingency tables

The gluing algebra forgets the geometry of couplings. A different algebra keeps it. Let f, g be rational margins on $\{1, \dots, n\}$ and $\{1, \dots, m\}$ with common denominator N, so that Nf and Ng are integer vectors with the same total. The polytope $P = N \cdot \mathrm{Cpl}(f, g)$ consists of the nonnegative real $n \times m$ tables with row sums Nf and column sums Ng. Its lattice is the set of integer tables, and the lattice points of kP are the integer tables with margins kNf and kNg.

Theorem 7.1 (The ring of contingency tables). *(i) P is a lattice polytope of dimension $(n-1)(m-1)$ when f, g have full support, and it has the integer decomposition property: every integer table in kP is a sum of k integer tables in P.*

(ii) The Ehrhart ring $S_P = \bigoplus_{k \geq 0} \mathbb{R}[kP \cap \mathbb{Z}^{n \times m}]$, graded by k with multiplication given by adding tables, is generated in degree one. Its Hilbert function $k \mapsto \dim S_{P,k}$ is the Ehrhart polynomial of P, of degree $(n-1)(m-1)$, with constant term 1 and leading coefficient the volume of P normalized to the lattice of integer circulations.

(iii) $\operatorname{Proj} S_P$ is the projective toric variety of P, embedded projectively normally. For nondegenerate margins it is the smooth toric variety $X_{f,g}$ of [3, Thm. 6.4], and S_P is the ring of sections of the powers of the ample line bundle of P, whose class in $\operatorname{Pic}(X_{f,g}) \otimes \mathbb{Q}$ is N times the class of the product coupling.

(iv) The degree-one basis $P \cap \mathbb{Z}^{n \times m}$ is the set of integer tables with margins Nf and Ng, the sample space of Fisher's exact test, on which the conditional law of a table given its margins under independence is the multivariate hypergeometric law, proportional to $1/\prod_{ij} t_{ij}!$. The basic moves $\delta_{ij} + \delta_{i'j'} - \delta_{ij'} - \delta_{i'j}$ connect this set, and each is an integer circulation.

Proof. (i) The constraint matrix of a transportation polytope is the incidence matrix of a complete bipartite graph, which is totally unimodular, so the vertices of P are integral [14], and polyhedra defined by totally unimodular systems with integral right-hand sides have the integer decomposition property [7]. The dimension is that of the circulation space. (ii) Generation in degree one is (i). The Hilbert function counts lattice points of kP, and the properties of the Ehrhart polynomial are standard [8]. (iii) A lattice polytope with the integer decomposition property is normal, hence very ample, and Proj of its Ehrhart ring is the toric variety of its normal fan, embedded projectively normally, and the sections of the k-th power of its line bundle have the lattice points of kP as a basis [10, Ch. 2 and Prop. 4.3.3]. The normal fan of P is that of $\mathrm{Cpl}(f, g)$, which for nondegenerate margins is the smooth fan of $X_{f,g}$. The support numbers of P are N times the entries of the product coupling, which give the class of $\mathrm{Cpl}(f, g)$ [3, Prop. 8.1]; an integral representative comes from any lattice point of P. (iv) The hypergeometric law is classical [1]. For two-way tables the basic moves form a Markov basis [11]; each has zero row and column sums. □

Example 7.2. For $f = (\frac{1}{2}, \frac{1}{2})$ and $g = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, take $N = 6$. The tables with margins $(3, 3)$ and $(2, 2, 2)$ are 7 lattice points of a hexagon. The numbers of tables with margins $k(3, 3)$ and $k(2, 2, 2)$ are $7, 19, 37, 61, 91$ for $k = 1, \dots, 5$, equal to $3k^2+3k+1$. The leading coefficient 3 is the lattice area of the hexagon, which is $N^2 = 36$ times the area $\frac{1}{12}$ of $\mathrm{Cpl}(f, g)$ in [3, §8]. The hexagon has one interior point, so it is the reflexive hexagon of the del Pezzo surface of degree 6, and its h^*-vector, read off from $\sum_k L(k)t^k = h^*(t)/(1-t)^3$, is $(1, 4, 1)$, of sum 6, the normalized area. For margins $(5, 7)$ and $(3, 4, 5)$ the counts give $10k^2 + 6k + 1$, the area of that polygon is 10, and $h^* = (1, 14, 5)$.

For two rows the dependence on the margins can be written down completely. Write $r = (r_1, r_2)$ and $c = (c_1, \dots, c_m)$ for integer margins with $r_1 + r_2 = \sum_j c_j$, $N(r, c)$ for the number of integer tables, and $c(S) = \sum_{j \in S} c_j$. The walls of [3, Def. 6.3] for two rows are the hyperplanes $r_1 = c(S)$ with $\emptyset \neq S \subsetneq \{1, \dots, m\}$, since $r_2 = c(S)$ is $r_1 = c(S^c)$.

Proposition 7.3 (Two-row tables across the walls)**.** *(i) $N(r, c) = \sum_S (-1)^{|S|} \binom{r_1 - c(S) - |S| + m - 1}{m-1}$, the sum over $S = \emptyset$ and the subsets $S \subseteq \{1, \dots, m\}$ with $c(S) < r_1$.*

(ii) On each chamber N is a polynomial in (r_1, c) of degree $m - 1$. Crossing the wall $r_1 = c(S)$ in the direction of increasing r_1 adds the polynomial $(-1)^{|S|} \binom{r_1 - c(S) - |S| + m - 1}{m-1}$, which vanishes on the lattice points with $|S| - m + 1 \leq r_1 - c(S) \leq |S| - 1$. So the polynomials of adjacent chambers agree on a strip along the wall.

(iii) For $m = 3$ and $|S| = 1$, the polygon on the far side is obtained by cutting a corner of lattice depth $d = r_1 - c_j$, and the count drops by $\binom{d+1}{2}$. This is the Riemann–Roch identity $\chi(\pi^* L - dE) = \chi(L) - \binom{d+1}{2}$ for the toric blow-up π at the corresponding fixed point with exceptional curve E, and its leading term $\frac{1}{2}d^2$ is, after rescaling by N, the area jump $\frac{1}{2}s^2$ of [3, Prop. 10.6(i)].

Proof. (i) A table is determined by its first row t, with $0 \leq t_j \leq c_j$ and $\sum t_j = r_1$. Without the upper bounds there are $\binom{r_1 + m - 1}{m-1}$ such rows, and inclusion–exclusion over the set S of violated bounds, substituting $t_j = c_j + 1 + t'_j$ for $j \in S$, gives the terms with $r_1 - c(S) - |S| \geq 0$. A term with $c(S) < r_1$ but $r_1 - c(S) - |S| < 0$ has top argument between 0 and $m - 2$, so it is zero, and including it changes nothing. (ii) On a chamber the set of subsets with $c(S) < r_1$ is constant, so (i) is a polynomial. Crossing $r_1 = c(S)$ adds exactly the term for S, and $\binom{x+m-1}{m-1}$ vanishes for $x = -1, \dots, -(m-1)$. (iii) For $m = 3$ the polygon is the triangle $\{t \geq 0,\ t_1 + t_2 + t_3 = r_1\}$ cut by the bounds $t_j \leq c_j$, and the term for $S = \{j\}$ is minus the number $\binom{d+1}{2}$ of lattice points with $t_j > c_j$, a corner triangle. For the toric surface, $(\pi^* L - dE)^2 = L^2 - d^2$ and $(\pi^* L - dE) \cdot K = L \cdot K + d$, since $K = \pi^* K + E$ and $E^2 = -1$; Riemann–Roch gives the identity, and the lattice points of the polygon count χ because the line bundle is nef. The leading term of $\binom{d+1}{2}$ is $\frac{1}{2}d^2$. □

General transportation polytopes behave the same way in outline: the number of tables is a vector partition function, polynomial on the chambers because the constraint matrix is unimodular [15], and its jumps across walls are given by the wall-crossing formula of Paradan as made explicit by Boysal and Vergne [9].

Example 7.4 (Fisher's exact test on synthetic data)**.** The first 40 items of the synthetic moderation data of [3, §12] (archived at https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/blob/main/coupling/synthetic_data.zip) give the label-by-action table

$$T = \begin{pmatrix} 10 & 8 & 1 \\ 2 & 5 & 14 \end{pmatrix}, \qquad \text{margins } (19, 21) \text{ and } (12, 13, 15).$$

Its fibre, the degree-one basis of the ring of $\frac{1}{40}\cdot$ these margins, has 151 tables. The hypergeometric probabilities sum to 1, and the exact p-value, the total probability of tables no more likely than T, is 9.8×10^{-5}. The χ^2 approximation with 2 degrees of freedom gives 1.8×10^{-4}.

The three appearances of the circulations now line up. They are the ideal I_f of the gluing algebra (section 2), the tangent space of the coupling polytope, whose dimension $(n-1)(m-1)$ is both the degree of the Ehrhart polynomial and the number of degrees of freedom of the test of independence [3, Prop. 12.1], and the lattice of Markov moves that connects the sample space of the exact test. The first appearance is homotopically trivial (theorem 5.2). The other two carry the geometry, because they keep positivity: a table must stay nonnegative under a move, and the polytope is cut out by nonnegativity.

8 Open problems

1. *The free cone at chain level.* In the free cone of I_f, products of homotopies survive, and the functor of theorem 5.1 acquires arity-four terms. Are the higher components given by the higher Boolean cumulants of whiskering? By theorem 5.2 any answer is invisible up to homotopy.

2. *A positive homotopy theory.* Corollary 5.3 locates contextuality in positivity: every linear invariant is blind to it, and every horn fills with signed measures [5, Prop. 35.1]. Find a homotopy theory of cones of measures, for example through simplicial distributions, in which unfillable horns of the coupling nerve [3, §4] are nontrivial classes. The concrete test case is the family of parity horns of [3, Thm. 4.3], unfillable with contextual fraction $1 - 2^{2-n}$, whose secondary obstruction in the tower is $o_3^- = 2\,\mathrm{CF}$ [3, Thm. 11.2]. A theory that sees them would be a positivity-preserving refinement of the square-zero cone: it must agree with the cone on signed measures, where it can only see masses, and it should recover the contextual fraction, or o_3^-, as the size of a class.

3. *Ehrhart data across walls.* Proposition 7.3 settles two rows. For $n \times m$ tables, compute the chamber polynomials and the h^*-vectors of $N \cdot \mathrm{Cpl}(f, g)$ chamber by chamber, and match the wall-crossing terms of [9] with the variation of GIT and the central-charge jumps of [3, §10] and [4, §3], as proposition 7.3(iii) does for one blow-up. This is the algebraic-statistical counterpart of the wall-crossing of the companion paper: the number of tables with given margins is the Euler characteristic of a line bundle, and its jumps should be those of the derived categories across the wall.

4. *Sections and potentials.* On $X_{f,g}$ the degree-one sections are indexed by tables. Relate the exact test to the disc potential of [3, §7], for example through the leximin coupling, which lies over the Floer-nontrivial fibres at the sampled points of [3, Obs. 10.10].

9 Computational checks

`check_coupling_algebra.py` runs fourteen checks, all passing:
- A1–A2: theorem 2.2 for $n = 2, \dots, 5$, and proposition 2.3 on objects of sizes $2, 3, 4$;
- A3: proposition 4.1;
- A4: theorem 5.1 for convex combinations of the two actions;
- A5: theorem 5.2;
- A6: proposition 6.1;
- A7–A9: theorem 7.1(i), (ii) and (iv) on two 2×3 examples and a 3×3 example;

- A10: example 7.4, in floating point;
- A11: proposition 3.1 for $n = 3, 4$, including the dimension of the span;
- A12: proposition 3.2, with an explicit level-two context;
- A13: proposition 3.3(ii), the sufficient condition of proposition 6.1 on both sides, and its failure for a generic merge;
- A14: proposition 7.3 against enumeration for $m = 3, 4$, the chamber polynomials and corner cuts, and the h^*-vectors of example 7.2.

Everything is exact rational arithmetic except the χ^2 tail in A10. The exact test itself is computed exactly: each hypergeometric probability is a rational number, and the 151 tables of the fibre are compared exactly. The χ^2 tail with two degrees of freedom is $e^{-X^2/2}$, evaluated in floating point.

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